



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Information Security Exposure and Data Loss



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.

Design the technological architecture of the IT system, through hardware and software recognition, based on available technology in the market, the analysis report and the distribution diagram.

Can Do Statements:

- I can explain how to avoid information security exposure.
- I can compare different strategies to avoid information security exposure.
- I can talk about information security exposure.

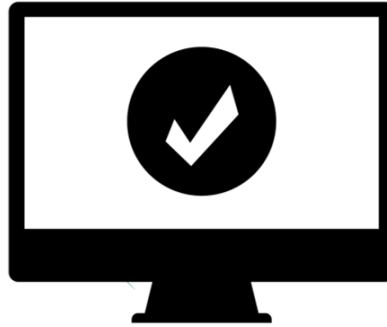
Ask your partner:

- Have you ever been hacked in any social network account? Explain the consequences that brought to you.
- What sensitive information have you ever lost in your studies or at work?



Let's Explore!

An IT student is talking to his professor about information security exposure.



Instructions: Read the text below.

- Student:** Professor, it's still not clear to me the difference between information security exposure and information system vulnerability. Could you help me note the difference?
- Professor:** Certainly. If the software of an information system doesn't have a defect in the programming code, hackers won't have the chance for direct access into the system to cause any harm or data loss, this is vulnerability. Now regarding information system exposure, unless we take proper care of the information even having a robust system, an attacker will steal or misuse it for fraudulent purposes.
- Student:** Oh, I see. So, inappropriate handling of information is a mistake that involves information security exposure. Is that right?
- Professor:** Exactly.
- Student:** Okay. And how could an IT officer avoid information security exposure?
- Professor:** One option is to promote strong password log-in into the system. If you don't set a robust password, hackers will have greater chances to access the confidential information. Another alternative lies on not guaranteeing access to everybody to sensitive data. If you guarantee former employees, who have a grudge against a company, complete access to the system without taking proper precautions, they might cause intentional harm or data loss. So, controlling accessibility is another strategy. Finally, proper data handling. For instance, unless you locate papers, a computer or any devices in the correct place, somebody will inevitably steal them and therefore the sensitive data contained there, too. Think of a laptop left at an airport or papers in a taxi, for example.
- Student:** Professor Clayton, which of these strategies would you say is the most effective?
- Professor:** I would not say there is one more effective than the others. Personally, I believe you should apply them together. If the IT officer does not promote and enforce good practices, the whole system sooner or later will be into trouble.



Let's Practice!

What will you do to avoid information security exposure?

Instructions: Find the words in the wordsearch below.

1. If you don't guarantee access to sensitive data to everybody, you will avoid information security _____.
2. We will install a more robust security _____ to avoid security exposure.
3. If you train your staff about the appropriate use of data and its _____, they will be more careful with the information management.
4. That's why it is so important to protect important _____.
5. That is why it is so important have a set of strict _____.

D	S	M	L	U	J	H	B	D	N	G	Y	Y	G	L
K	Y	S	H	L	R	L	N	G	R	A	A	U	K	G
S	S	C	A	B	I	N	I	S	B	X	T	D	K	J
K	T	P	N	Q	F	G	X	Y	C	N	U	A	B	A
Q	E	C	N	O	Y	T	O	S	A	T	D	T	Z	K
T	M	V	E	X	P	O	S	U	R	E	P	A	R	E
Q	E	F	T	K	E	A	M	E	D	B	R	I	J	D
C	T	S	N	M	V	G	G	M	K	L	K	L	D	M
A	J	A	A	F	I	D	G	A	T	J	F	L	B	P
N	D	P	R	O	T	E	C	T	I	O	N	E	T	H
N	N	R	G	D	W	A	M	P	I	B	A	G	B	S
Y	B	R	E	A	D	E	R	P	V	G	H	E	U	P
B	B	N	R	W	Z	B	D	K	V	N	V	S	S	Y
O	Y	M	U	P	A	S	S	W	O	R	D	S	L	J
M	N	D	V	L	K	F	A	U	O	N	R	N	C	Y



Let's Practice!

What will happen if we don't avoid information security exposure?

Instructions: Match column A with column B.

COLUMN A

1. There will be a risk if the software of an information system
2. If there isn't any vulnerability, hackers won't have
3. If we don't take good care of information,

COLUMN B

- a. somebody might steal it and use it for bad purposes.
- b. doesn't work properly.
- c. direct access to the system.





Extension Activity

I look for the best strategy to avoid information security exposure.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about information security exposure.
- Identify different strategies to avoid information security exposure.
- Write down a paragraph mentioning which are the best strategies and in which conditions you will use those strategies.
- Share your paragraph with the class and discuss about it.



Target Words

- **IT officer** (noun): title given to someone responsible for the company's information technology strategy and supervision of the people, technology and processes associated to it.
e.g. If the IT officer does not promote and enforce good practices, the whole system sooner or later will be into trouble.
- **password** (noun): a personal and secret series of words and numbers used to guarantee individual access to a system, computer or program
e.g. One option is to promote strong password log-in into the system.
- **enforce** (verb): to impose or make other people obey rules or policies or to be accepted
e.g. If the IT officer does not promote and enforce good practices, the whole system sooner or later will be into trouble.

Self-Assessment

Yes

No

- I search for information about information security exposure.
- I identify different strategies to avoid information security exposure.
- I write about the best strategies to avoid information security exposure and in which conditions I will use those strategies.
- I share information and discuss about the topic.

☐	☐
☐	☐
☐	☐
☐	☐

- **grudge** (noun): a feeling resentment and dislike felt for someone who has treated you badly
e.g. If you guarantee former employees, who have a grudge against a company, complete access to the system without taking proper precautions, they might cause intentional harm or data loss.
- **programming code** (noun): the fundamental component of a programming language that consists of written instructions used to make a machine perform a task.
e.g. Certainly. If the software of an information system doesn't have a defect in the programming code, hackers won't have the chance for direct access into the system to cause any harm or data loss, this is vulnerability.





Learning Points

First conditional with the if clause in negative

There are some types of conditions in English. Some are possible, others are unlikely while others are impossible. **The first conditional** is about a real condition with a possible or imagined future consequence or result.

Form the **first conditional** with two clauses. One of them is the likely condition and the other clause is the imagined future consequence or result. It doesn't matter which clause you use first, but in case you begin with the conditional clause, you must place a comma at the end of the first sentence before you can begin the second sentence.

Examples:

- **If we enforce** employees to follow the rules for the correct use of our information system, it **will** reduce complaints and petitions over 40% in a period of just three months.
- Enforcing employees to follow the rules **will** reduce complaints and petitions over 40% in a period of just three months **if** we **do** it for the correct use of our information system.

Conditional with if in negative clause

You use the word "if" to state that the condition is likely to happen in opposition to the word "when" that means the condition is more definite. In the negative form, you can express the condition with if in another way:
If not = unless

Examples:

- **If we don't take** the necessary precautions, the system might collapse for data stolen or data corruption.
- **Unless we take** the necessary precautions, the system might collapse for data stolen or data corruption.
- The system **might** collapse for data stolen or data corruption **if** we **don't take** the necessary precautions

Grammar structure

The grammatical structure for the first conditional is pretty straightforward.

If/unless + subject + don't/doesn't + verb (present) + complement, (Clause 1)

subject + will/may/might/can + complement (Clause 2)

Example:

- **If we don't check** the security policies and guidelines, end users sooner than later **will complain** for the improper functioning of the system.
- **Unless we** reassign authorized access to some of our employees, we **will keep** experiencing data loss and data corruption.





Learning Points

Examples

- *If the software of an information system doesn't have a defect in the programming code, hackers won't have the chance for direct access into the system to cause any harm or data loss, this is vulnerability.*
- *For instance, unless you locate papers, a computer or any devices in the correct place, somebody will inevitably steal them and therefore the sensitive data contained there, too.*

Best network security programs

Network security programs are tools specially designed to prevent data loss, reduce the system's vulnerability and keep the integrity, availability and confidentiality of the information.

There are many network security programs available in the market used to protect an information system. Here we'll describe briefly some of the functionalities and capabilities of such programs.

Depending on the robustness and complexity of the program, they can reduce cyber-attacks when they:

- Perform log aggregation.
- Perform log analysis.
- Do compliance reporting.
- Realize file integrity monitoring.
- Monitor user activity.
- Perform event correlation.
- Provide real-time alerting.
- Perform log forensics and log retention.
- Identify root causes of security incidents and data breaches.
- Also, offer email encryption and email delivery proof.
- Protect against third party spying.
- Block access to restricted websites and isp speed throttling.
- Protect users against malicious programs such as worms, trojans, viruses, hackers, spam and other ways of intrusion.

